MISSING DATA & STATISTICAL INFERENCE

A Simulation Study of MCAR, MAR and MNAR

Comparing Eight Ways of Handling Missing Values
Within Each Missingness Mechanism

Project Report
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Summary

Real datasets very often have missing values. This project asks a simple question: **if some values are missing, does the way we deal with them change our results, and can we still trust the confidence interval that comes with them?**

To answer it, we created data from a model we know, $Y = 5 + 2X_1 - 1.5X_2 + X_3 + \text{error}$, with 200 observations. Because we know the model, we know the true value of the slope we care about, $\beta_1 = 2$. We then deleted about 20% of the values of Y and of X_1 in three different ways: completely at random (MCAR), in a way that depends on another variable that is still observed (MAR), and in a way that depends on the value that disappears (MNAR). To every incomplete dataset we applied eight methods: complete-case analysis, mean imputation, hot-deck imputation, regression imputation, stochastic regression imputation, two versions of multiple imputation (MICE), and the EM algorithm.

The study has two parts. In **Part A** one dataset is analysed step by step. In **Part B** the whole experiment is repeated 300 times with fresh data, so that we can see how each method behaves *on average*. All comparisons are made **within** each mechanism: for MCAR, then MAR, then MNAR, we ask which method works best.

Main findings

- **Mean imputation is the worst method in every setting.** Its bias is -0.40, -0.47 and -0.55 (MCAR, MAR, MNAR), and its 95% intervals contain the true value in only 12%, 5% and 6% of the runs.
- **Hot-deck, regression imputation and stochastic regression report standard errors that are too small** (roughly 70–85% of the true spread), so their intervals are too narrow. They treat made-up values as if they were real data.
- **MCAR:** complete-case analysis works well; MICE and EM are slightly more precise.
- **MAR:** complete-case analysis is biased and its intervals cover the truth only 80% of the time. EM and MICE (normal) are almost unbiased with about 95% coverage, because they use an extra observed variable.
- **MNAR:** no method fixes the problem. Even the best intervals cover the truth only 89–91% of the time.

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1 Introduction

Missing data are very common. A survey respondent skips a question, a machine fails to record a reading, a patient leaves a study early. When values are missing, the analysis changes for two reasons. First, there is less information. Second, the way we *handle* the missing values can change the relationships we find and how sure we are about them.

How serious the problem is depends on *why* the values are missing. Statisticians describe three situations:

- **MCAR (Missing Completely At Random).** Whether a value is missing has nothing to do with the data. Example: a sensor fails at random moments.
- **MAR (Missing At Random).** Whether a value is missing depends on something we *did* observe. Example: older people skip a question more often, and age is recorded for everyone.
- **MNAR (Missing Not At Random).** Whether a value is missing depends on the missing value itself. Example: people with very high incomes refuse to report their income.

These three form a ladder: MCAR is the simplest case, MAR is more general, and MNAR is the most general (MCAR $\subset$ MAR $\subset$ MNAR).

There are many ways to deal with missing values. We can delete the incomplete rows, fill the gaps with an average, copy values from similar rows, predict them with a regression, create several completed datasets, or estimate the model directly with likelihood methods. These choices can lead to different estimates *and* to different confidence intervals. This project compares them in a controlled experiment where the true answer is known.

2 Aim and Research Question

2.1 Aim

The aim is to find out how different missing-data methods affect a regression estimate and its uncertainty, when all methods are given the same incomplete data from the same missingness mechanism. We look at both the **estimate** itself and the **confidence interval** around it. The regression has four coefficients, but we focus on β_1, the slope of X_1, whose true value is known to be $\beta_1 = 2$.

2.2 Research question

The question. *Within each type of missingness, which way of handling the missing values gives the most accurate estimate and the most honest confidence interval?* An "honest" interval is one that contains the true value about 95% of the time, as a 95% interval should.

The question is answered three times: once for MCAR, once for MAR and once for MNAR. We do not rank the three mechanisms against each other. Each one is a separate setting in which the eight methods are compared.

3 How the Data Were Created

A complete dataset was generated first, before any value was deleted. The outcome follows the linear model

$$Y = 5 + 2X_1 - 1.5X_2 + 1X_3 + \varepsilon,$$

where X_1, X_2, X_3 are independent standard normal variables and the error ε is standard normal. Every dataset has **n = 200** observations. Because we chose the coefficients, the true values are known: 5, 2, −1.5 and 1.

We also create an **auxiliary variable** $Z = Y +$ noise. It is a noisy copy of Y (their correlation is about 0.94). Z is **not** part of the regression and is never missing, but it can help to fill in the gaps. Think of it as a cheap, rough measurement of the same thing, such as a self-reported value next to an exactly measured one.

3.1 Baseline: the regression with no missing data

Before deleting anything, the regression was fitted to the complete data. This is the best any method could hope to match.

Parameter	True value	Estimate from the complete data	Standard error
β0 (intercept)	5.000	4.9629	0.0726
β1	2.000	2.0368	0.0722
β2	-1.500	-1.5076	0.0749
β3	1.000	0.9483	0.0776

Table 1: Baseline regression on the complete data (no missing values).

The complete-data estimate of β_1 is 2.037, close to but not exactly 2. This is normal: a dataset is only a sample, so even with no missing data the estimate has some random error. It is important to keep this in mind when reading the results: not every difference from 2 is caused by missing data.

4 How Values Were Deleted

About **20% of the values of *Y* and about 20% of the values of X_1** were deleted, each one separately. Because two variables lose values, roughly a third of the rows have at least one gap. The three mechanisms differ only in *how likely* a value is to be deleted (Figure 1).

- **MCAR:** every value has the same 20% chance of being deleted. It is pure luck.
- **MAR:** the chance of deletion depends on Z, which is always observed. Rows with a large Z lose their values more often.
- **MNAR:** the chance of deletion depends on the value that gets deleted. Large Y values and large X_1 values vanish more often.

For MAR and MNAR the chance of deletion is expit$(-2.23 + 2 \times d)$, where d is the standardised “driver” (Z for MAR, the value itself for MNAR) and expit is the S-shaped

curve that turns any number into a probability between 0 and 1. The number −2.23 was tuned so that about 20% of values are deleted in every mechanism.

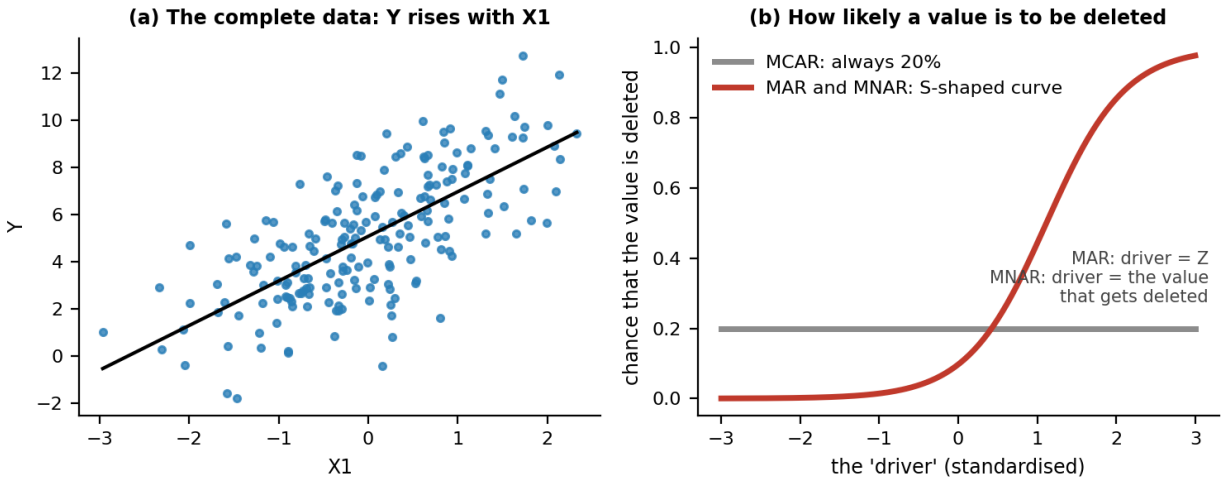


Figure 1: (a) The complete data: Y increases with X1. (b) The chance that a value is deleted under each mechanism.

Table 2 shows how many values were actually deleted in the one dataset used in Part A. The numbers are not exactly 20% because deletion is random.

Mechanism	X1 missing	Y missing	Rows with at least one gap	Complete rows left
MCAR	45	43	78 (39%)	122
MAR	39	40	61 (30%)	139
MNAR	41	41	65 (32%)	135

Table 2: Missing values in the dataset used in Part A (out of 200 rows).

We also checked that the rules behave as intended, using a very large dataset (20,000 rows) so that random noise does not hide the pattern (Table 3). Under MCAR, rows with a missing Y look just like the others. Under MAR and MNAR, rows with a missing Y have clearly higher Z and higher Y.

Mechanism	Y missing (%)	X1 missing (%)	Rows with a gap (%)	Z: missing minus observed	Y: missing minus observed
MCAR	20.11	20.02	36.10	-0.01	-0.03
MAR	19.87	20.16	30.58	3.96	3.53
MNAR	19.82	20.08	32.28	3.74	3.72

Table 3: Check of the three deletion rules on 20,000 rows.

In simple words. In real data we can see Z but we cannot see the deleted Y values, so MAR and MNAR cannot be told apart from the observed data alone. We can only tell them apart here because we deleted the values ourselves.

5 The Eight Methods

Every method ends in the same step: a regression of Y on X_1, X_2 and X_3. The methods differ only in what they do with the gaps before that regression.

Method	How it works	Good	Not so good
Complete case	Delete every row with a gap.	Simple. Fine under MCAR.	Throws away rows; can be biased otherwise.
Mean imputation	Fill each gap with the column average.	Very simple.	Shrinks the spread and weakens relationships.
Hot-deck	Copy the value from the most similar row.	Uses real values.	One guess per gap; treated as if real.
Regression imputation	Fill each gap with the value predicted by a regression.	Uses the other variables.	Filled values sit exactly on the line.
Stochastic regression	Predicted value plus a random residual.	Restores natural scatter.	Still one guess per gap.
MICE (normal)	Make 10 random completed datasets, analyse each, combine.	Honest uncertainty.	More work.
MICE (PMM)	Same, but borrow real values from similar rows.	Honest uncertainty; real values.	Needs enough similar donors.
EM	Maximum likelihood using every row, even partly observed ones.	Uses all information.	Assumes a normal distribution.

Table 4: The eight methods at a glance.

5.1 Complete-case analysis

Rows with any missing value are removed and the regression is fitted to the rest. No made-up values are created, but information is thrown away: in this study between 120 and 140 of the 200 rows remain.

5.2 Mean imputation

Each gap is filled with the average of the observed values in that column. All filled values are identical, so the spread of the data shrinks and the link between Y and X_1 becomes weaker.

5.3 Hot-deck imputation

For each row with a gap, the most similar row (the nearest neighbour, judged on the variables that are observed in both rows, after putting them on the same scale) is found, and its value is copied. The filled values are real observed values, but each gap gets only one guess.

5.4 Regression imputation and stochastic regression

Each incomplete variable is regressed on all the other variables, Z included, and the variables are filled in one at a time, cycling through them five times. In **regression imputation** the gap receives the predicted value. In **stochastic regression** a randomly chosen residual from the fitted regression is added, so the filled values scatter around the line the way real data do.

5.5 MICE: multiple imputation

MICE (Multiple Imputation by Chained Equations) does not fill each gap just once. It creates **m = 10** different completed datasets. Each time, new regression coefficients are drawn first, so the uncertainty about the model itself is included. In the **normal** version the gap is filled with the prediction plus a random draw. In the **PMM** version (predictive mean matching) the five rows whose predicted values are closest are found, and the real observed value of one of them is copied.

The regression is then run on every completed dataset and the 10 results are combined with **Rubin's rules**. The estimate is the average of the 10 estimates. The total variance is

$$\text{total} = \text{within} + (1 + 1/m) \times \text{between},$$

where *within* is the average variance inside each completed dataset and *between* measures how much the 10 estimates disagree. The *between* part captures the extra uncertainty caused by the missing data. A method that fills each gap only once has no such part.

5.6 EM algorithm

EM estimates the means and covariances of all five variables by maximum likelihood, using every row, including partly observed ones. It repeats two steps until nothing changes: the **E-step** fills each gap with its expected value given what is observed, and the **M-step** re-estimates the means and covariances. The regression coefficients are then calculated directly from them. EM has no simple formula for its standard error, so a bootstrap is used.

5.7 What the methods do to the data

Figure 2 shows Y against X_1 in the MAR dataset after each method has handled the gaps. Orange points are rows where a value was missing. The black line is the fitted line, with its slope printed above each panel. (This is the slope of Y on X_1 alone, so for the full data it is 1.89 rather than the 2.037 of the full regression; compare the panels with each other.)

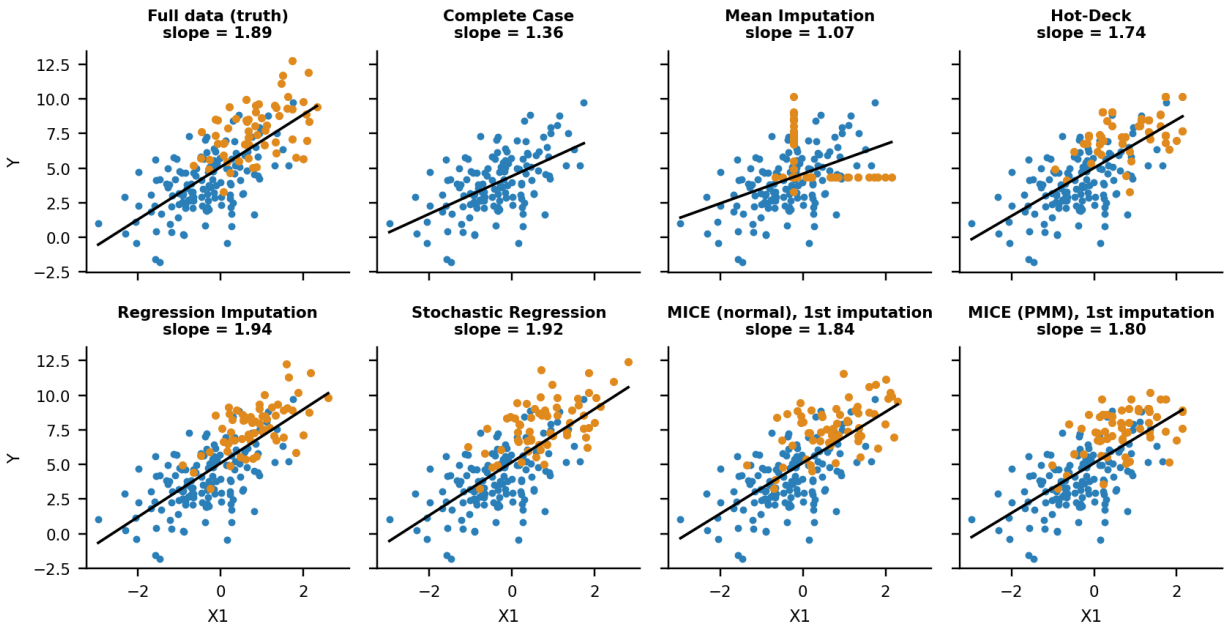


Figure 2: What each method did to the data (MAR dataset). Orange = rows where a value was missing.

Complete case loses the top-right part of the cloud (rows with large Z are the ones that lose values under MAR), so its slope is too flat. Mean imputation puts the filled points on a horizontal and a vertical line through the average, which flattens the slope a lot. Regression imputation places the filled points exactly on the fitted line. The remaining methods scatter the filled points around the line like real data.

6 How the Methods Are Judged

6.1 Part A: one dataset

In Part A each method is applied once to each of the three incomplete datasets. For β_1 we report the estimate, its **bias** (estimate − 2), a standard error and a 95% confidence interval. Two kinds of difference are shown:

- **Bias vs truth** = estimate − 2. This mixes the damage done by missing data with ordinary random error.
- **Difference from the full-data estimate** = estimate − 2.037. This isolates what the missing data did.

The standard error is found in one of two ways. For every method except MICE we use the **bootstrap**: the rows are resampled with replacement 500 times, the *whole* method (including the filling-in) is repeated each time, and the spread of the 500 estimates gives the standard error. The middle 95% of the 500 estimates gives the interval. For MICE we use Rubin's rules. We also show the **naive SE**, which is what you get by running the regression on the completed data and forgetting that some values were made up.

6.2 Part B: 300 repetitions

One dataset can mislead, because one dataset is one random draw. In Part B the whole experiment is repeated **300 times**: each time a new complete dataset is generated, damaged under each mechanism, and analysed with every method. We also analyse the same data with *no* missing values as a benchmark. The measures are in Table 5.

Measure	What it tells us
Bias	Average of (estimate − 2). Zero means right on average.
Spread (SD)	How much the estimate varies from one dataset to the next: the true uncertainty.
RMSE	Overall accuracy; combines bias and spread. Smaller is better.
Reported SE	The standard error the method itself reports. If it is clearly smaller than the spread, the method is over-confident.
Coverage (%)	Share of the 300 intervals (estimate ± 1.96 × SE) that contain the true value 2. It should be about 95.

Table 5: Measures used in the simulation study.

With 300 repetitions, a coverage figure near 95% has a random error of about 1.3 percentage points, so differences of 2–3 points or less should not be over-read. How each method gets its standard error inside the simulation is listed in Appendix B.

Benchmark. With no missing data, the spread of the estimate is 0.070 and the intervals cover the truth 95% of the time. No method can beat this benchmark. It also shows that the simulation itself works as it should.

7 Results: Part A (One Dataset)

The complete-data estimate for this dataset is 2.037. Tables 6 to 8 show every method under each mechanism, and Figure 3 shows the same intervals.

Method	β1 estimate	Naive SE	Final SE	95% interval	Bias vs truth	Diff. from full data
Complete Case	2.098	0.092	0.094	(1.899, 2.265)	+0.098	+0.061
Mean Imputation	1.622	0.126	0.146	(1.332, 1.898)	-0.378	-0.415
Hot-Deck	2.005	0.082	0.124	(1.740, 2.221)	+0.005	-0.032
Regression Imputation	2.168	0.060	0.082	(2.012, 2.318)	+0.168	+0.131
Stochastic Regression	2.060	0.069	0.089	(1.898, 2.257)	+0.060	+0.023
MICE (normal)	2.082	0.082	0.082	(1.922, 2.241)	+0.082	+0.045
MICE (PMM)	2.062	0.077	0.077	(1.912, 2.212)	+0.062	+0.025
EM	2.076	–	0.083	(1.916, 2.230)	+0.076	+0.040

Table 6: Part A, MCAR: all eight methods on the same dataset.

Method	β1 estimate	Naive SE	Final SE	95% interval	Bias vs truth	Diff. from full data
Complete Case	1.844	0.105	0.102	(1.629, 2.053)	-0.156	-0.193
Mean Imputation	1.346	0.133	0.137	(1.065, 1.585)	-0.654	-0.691
Hot-Deck	1.878	0.084	0.120	(1.599, 2.063)	-0.122	-0.159
Regression Imputation	2.110	0.068	0.096	(1.924, 2.298)	+0.110	+0.074
Stochastic Regression	2.085	0.079	0.099	(1.857, 2.231)	+0.085	+0.048
MICE (normal)	2.036	0.098	0.098	(1.843, 2.228)	+0.036	-0.001
MICE (PMM)	1.943	0.099	0.099	(1.749, 2.137)	-0.057	-0.093
EM	2.022	–	0.093	(1.830, 2.197)	+0.022	-0.015

Table 7: Part A, MAR: all eight methods on the same dataset.

Method	β1 estimate	Naive SE	Final SE	95% interval	Bias vs truth	Diff. from full data
Complete Case	2.015	0.115	0.103	(1.834, 2.229)	+0.015	-0.022
Mean Imputation	1.263	0.151	0.167	(0.946, 1.599)	-0.737	-0.774
Hot-Deck	1.935	0.111	0.124	(1.692, 2.169)	-0.065	-0.102
Regression Imputation	2.212	0.071	0.091	(2.049, 2.394)	+0.212	+0.175
Stochastic Regression	2.120	0.079	0.101	(1.913, 2.311)	+0.120	+0.084
MICE (normal)	2.107	0.082	0.082	(1.945, 2.268)	+0.107	+0.070
MICE (PMM)	2.047	0.111	0.111	(1.830, 2.264)	+0.047	+0.010
EM	2.108	–	0.088	(1.937, 2.287)	+0.108	+0.071

Table 8: Part A, MNAR: all eight methods on the same dataset.

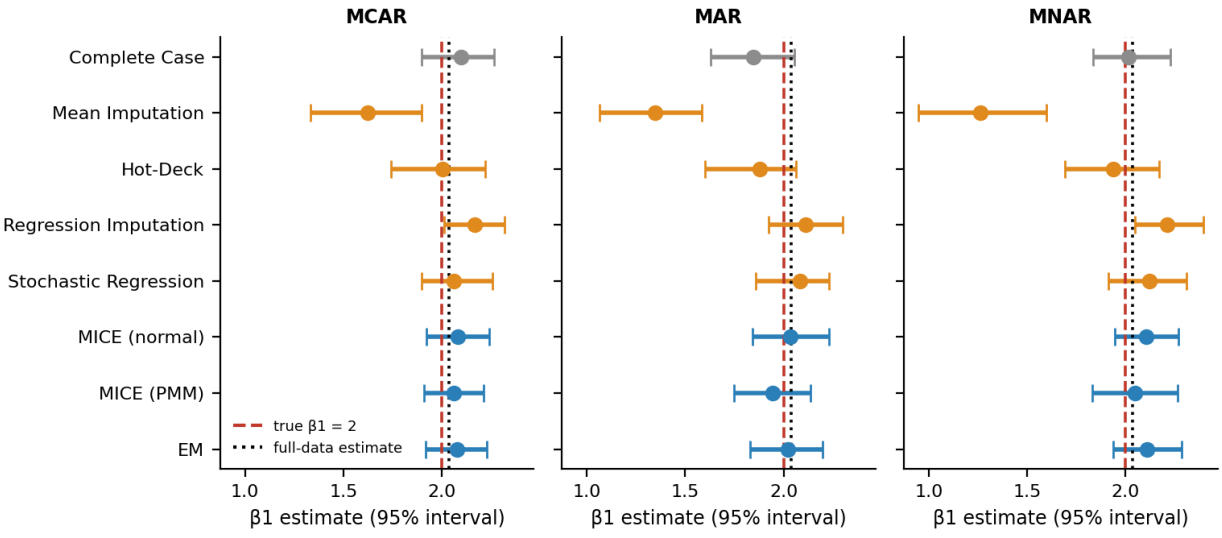


Figure 3: Part A. β1 estimate with 95% interval for each method. Red dashed line = true value 2; dotted line = full-data estimate.

7.1 What Part A shows

- **Mean imputation is far below the full-data estimate** in all three settings (1.62, 1.35 and 1.26, compared with 2.04).
- **The naive SE is smaller than the final SE for hot-deck, regression imputation and stochastic regression** in all three settings. For example, hot-deck under MCAR has a naive SE of 0.082 but a bootstrap SE of 0.124. Ignoring that values were made up makes the method look more certain than it is.
- **For MICE the naive and final SEs are the same**, because Rubin's rules already include the extra uncertainty.
- **One dataset can mislead.** Complete case looks fine under MNAR here (2.015), yet Part B shows it is biased on average. This is why a simulation study is needed.

8 Results: Part B (300 Repetitions)

For each mechanism we show a table ranked by RMSE (best first) and a figure with three panels: (a) the spread of the 300 estimates, (b) the coverage of the 95% interval, and (c) the reported SE divided by the true spread. The colours group the methods: grey = delete incomplete rows, orange = single imputation, blue = multiple imputation and EM.

8.1 MCAR: the gaps are pure luck

Rank	Method	Bias	Spread (SD)	RMSE	Reported SE	Coverage (%)
1	MICE (normal)	-0.005	0.080	0.080	0.083	95.7
2	EM	-0.001	0.080	0.080	0.082	94.3
3	MICE (PMM)	-0.011	0.082	0.083	0.083	95.7
4	Complete Case	+0.001	0.085	0.085	0.089	96.0
5	Stochastic Regression	-0.003	0.092	0.091	0.071	85.7
6	Regression Imputation	+0.081	0.081	0.114	0.062	68.0
7	Hot-Deck	-0.094	0.098	0.136	0.084	74.3
8	Mean Imputation	-0.397	0.133	0.419	0.125	12.3

Table 9: Part B, MCAR: methods ranked by RMSE (300 repetitions).

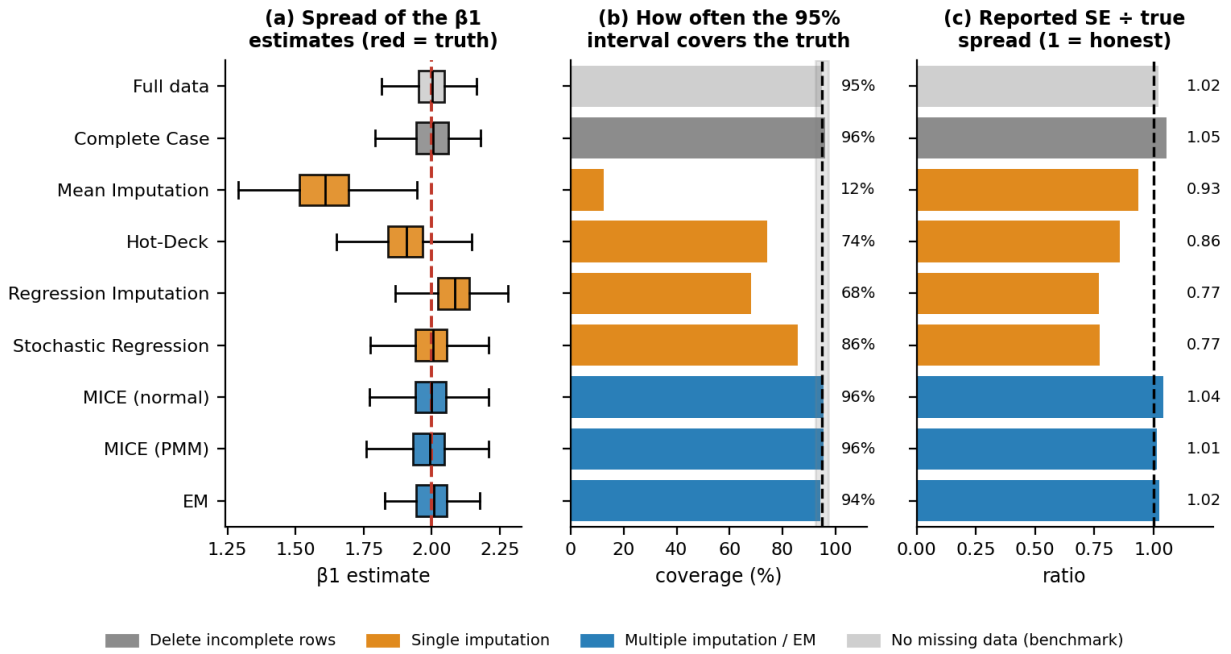


Figure 4: MCAR. Spread of the estimates, coverage of the 95% interval, and honesty of the reported SE.

- **Complete case is fine.** It is right on average (bias +0.001) and its intervals cover the truth 96% of the time. Its only cost is precision, because it uses fewer rows.
- **MICE (normal), MICE (PMM) and EM** have about the same accuracy as each other (RMSE 0.080, 0.083 and 0.080), slightly better than complete case (0.085), with coverage close to 95%.
- **Single imputation fails even though the data are MCAR.** Mean imputation is strongly biased (-0.40), regression imputation is biased upward (+0.08) and hot-deck downward (-0.09). Stochastic regression is almost unbiased (-0.003), but its coverage is only 86%, because its reported SE is too small.

8.2 MAR: the gaps depend on an observed variable

Rank	Method	Bias	Spread (SD)	RMSE	Reported SE	Coverage (%)
1	EM	+0.002	0.085	0.085	0.086	95.3
2	MICE (normal)	-0.001	0.086	0.086	0.087	95.3
3	Stochastic Regression	+0.006	0.092	0.092	0.072	86.3
4	MICE (PMM)	-0.052	0.097	0.110	0.091	87.7
5	Regression Imputation	+0.084	0.087	0.121	0.065	71.3
6	Complete Case	-0.100	0.093	0.136	0.092	79.7
7	Hot-Deck	-0.147	0.124	0.192	0.089	57.7
8	Mean Imputation	-0.471	0.130	0.489	0.128	5.3

Table 10: Part B, MAR: methods ranked by RMSE (300 repetitions).

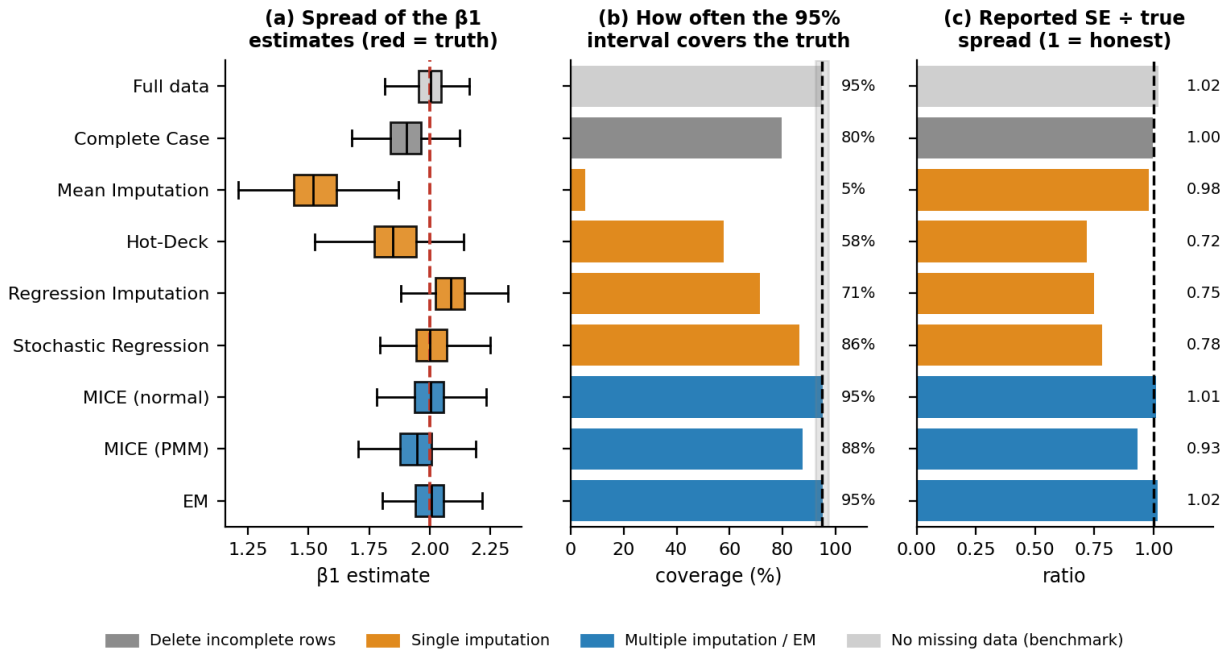


Figure 5: MAR. Spread of the estimates, coverage of the 95% interval, and honesty of the reported SE.

- **Complete case is no longer safe.** It is biased downward (-0.100) and its coverage falls to 80%. The rows with a missing Y are not a random sample: they tend to have large Z, which goes with large Y.
- **EM and MICE (normal) fix this.** They use Z, are almost unbiased (+0.002 and -0.001), cover 95% and 95% of the time, and have much lower RMSE than complete case (0.085 and 0.086 against 0.136).
- **MICE (PMM)** has a small downward bias (-0.052) and lower coverage (88%).
- **Single imputation again under-covers.** Stochastic regression is unbiased (+0.006) but covers only 86%. Hot-deck (58%) and regression imputation (71%) are biased as well. Mean imputation is by far the worst (5%).

Figure 6 makes coverage concrete. It shows the first 40 of the 300 intervals for four methods. Green intervals contain the true value 2 and red ones miss it. Complete case misses it too often because its intervals are centred too low. Stochastic regression is centred correctly but its intervals are too short.

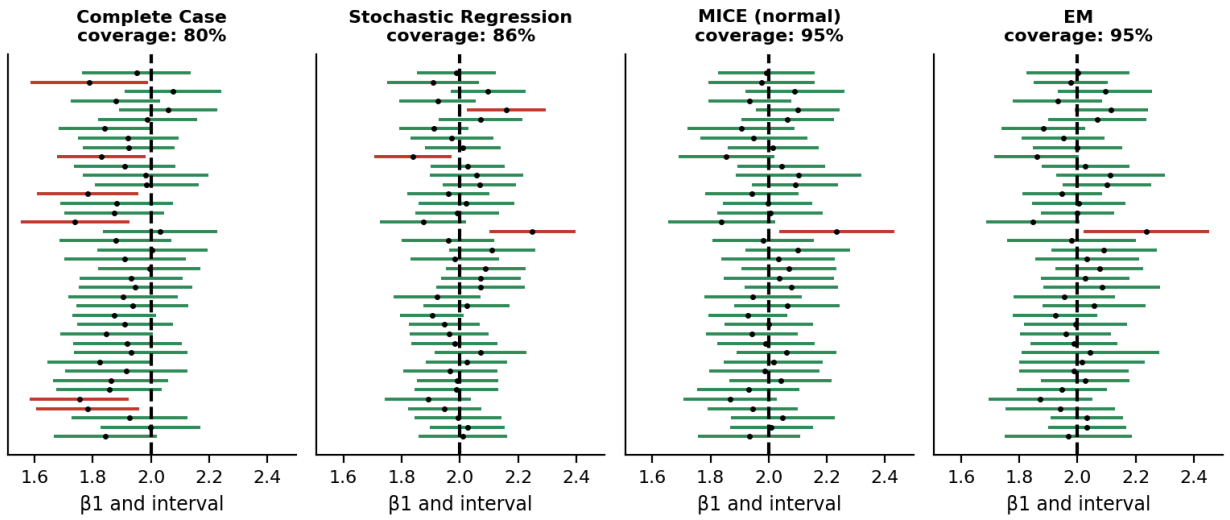


Figure 6: MAR. The first 40 confidence intervals of four methods (green = contains the truth, red = misses it).

8.3 MNAR: the gaps depend on the missing value itself

Rank	Method	Bias	Spread (SD)	RMSE	Reported SE	Coverage (%)
1	MICE (PMM)	+0.016	0.112	0.113	0.098	91.3
2	MICE (normal)	+0.063	0.096	0.115	0.094	89.7
3	EM	+0.066	0.095	0.116	0.092	88.7
4	Complete Case	-0.061	0.106	0.123	0.102	90.7
5	Stochastic Regression	+0.069	0.103	0.124	0.078	81.3
6	Hot-Deck	-0.046	0.128	0.136	0.096	84.7
7	Regression Imputation	+0.154	0.099	0.183	0.070	43.3
8	Mean Imputation	-0.547	0.156	0.569	0.147	5.7

Table 11: Part B, MNAR: methods ranked by RMSE (300 repetitions).

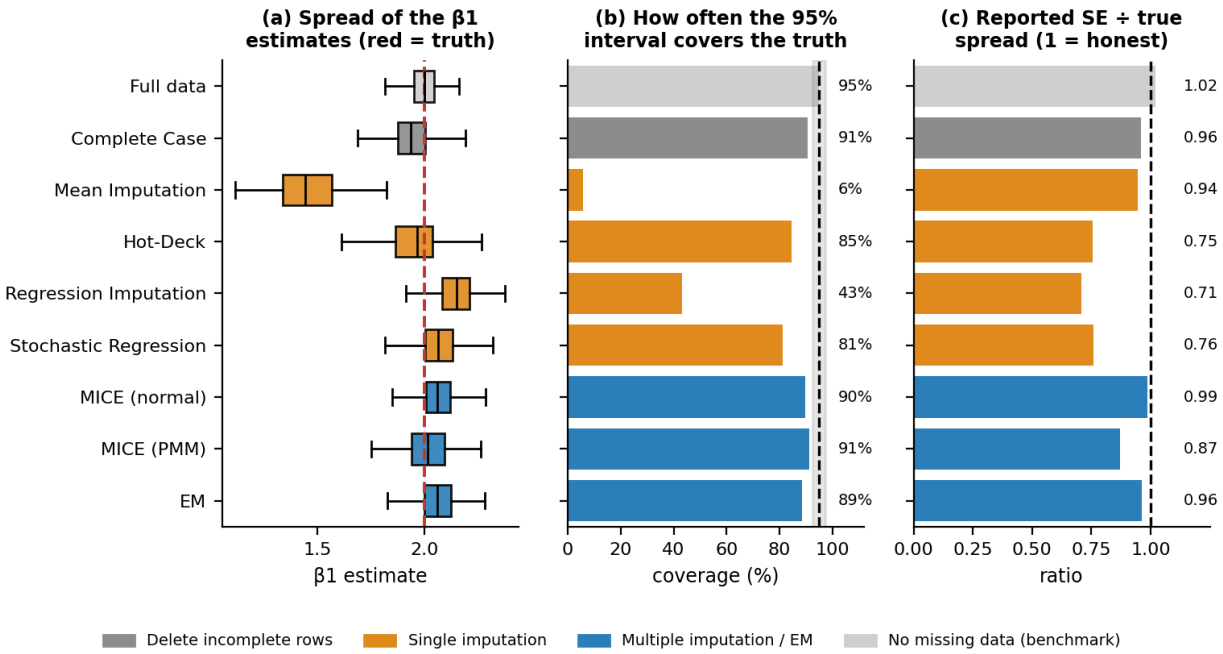


Figure 7: MNAR. Spread of the estimates, coverage of the 95% interval, and honesty of the reported SE.

- **No method is reliable.** Complete case is biased downward (-0.061), while EM, MICE (normal) and stochastic regression, which all assume the data are MAR, overshoot upward (+0.066, +0.063, +0.069). Even the best intervals cover only 89–91% of the time.
- **The leading methods are hard to separate.** Complete case, MICE (normal), MICE (PMM) and EM have similar RMSE (0.113–0.123), within random error of each other. The small bias of MICE (PMM) is probably two errors that happen to cancel, not a sign that it handles MNAR.
- **Regression imputation** (coverage 43%) and **mean imputation** (coverage 6%) are clearly the worst.

8.4 At a glance

Figure 8 puts all the results on one page. Each row is a method and each column a mechanism, so you can read down a column to compare the methods under one mechanism. Table 12 lists the best method for each mechanism according to several criteria.

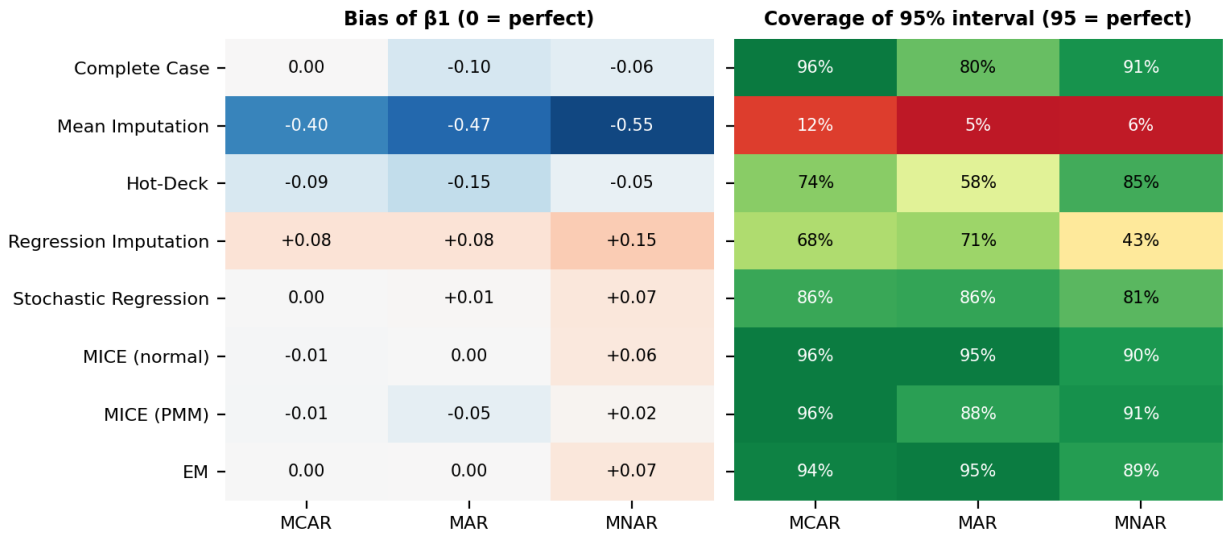


Figure 8: Bias (left) and coverage (right) of every method under every mechanism.

Mechanism	Lowest RMSE	Smallest bias	Coverage closest to 95%	Lowest RMSE among methods with coverage ≥ 90%
MCAR	MICE (normal)	Complete Case	EM	MICE (normal)
MAR	EM	MICE (normal)	MICE (normal)	EM
MNAR	MICE (PMM)	MICE (PMM)	MICE (PMM)	MICE (PMM)

Table 12: The best method under each mechanism, by four criteria.

Where the leading methods are within random error of each other (see the notes above), the “best” in this table should not be taken as a real difference.

9 Discussion

9.1 Mean imputation hurts everywhere

Filling every gap with the same number removes the natural variation in Y and X_1, and with it part of the relationship between them (Figure 2). The slope is pulled towards zero. This is not a small effect: the bias is between -0.40 and -0.55 in all three settings, and the interval almost never contains the truth. It does not matter whether the data are MCAR, MAR or MNAR.

9.2 Why single imputation is over-confident

Hot-deck, regression imputation and stochastic regression fill each gap once and then treat the result as real data. But about a third of the rows contain made-up values, so the data look more certain than they are. The reported SE is only about 70–85% of the true spread (panel (c) of Figures 4, 5 and 7), and the intervals are too narrow. Stochastic regression is the clearest example: under MCAR and MAR it has almost no bias, yet its 95% intervals contain the truth only about 86% of the time.

In simple words. A good estimate is not enough. A method can get the right answer on average and still give you a confidence interval you should not believe.

9.3 What multiple imputation and EM do differently

MICE (normal) uses almost the same filling-in model as stochastic regression. It adds only three things: new coefficients are drawn each time, the filling-in is repeated 10 times, and Rubin's rules combine the results. These changes are enough to bring the coverage from clearly below 95% up to about 95% under MCAR and MAR. EM reaches the same accuracy by a different route, by using the likelihood of all the observed data.

9.4 Why complete case fails under MAR here, and the role of Z

Under MAR the rows that lose values are those with a large Z. Since Z is closely tied to Y, the complete rows are a biased selection of the data. Methods that use Z in the filling-in step (EM and MICE) can correct for this. Complete case cannot, because it ignores Z. The lesson: **variables that explain why data are missing belong in the imputation model, even if they are not in the analysis model.** In this study Z is a very good stand-in for Y; in real data the auxiliary variables are usually weaker, so the advantage would be smaller.

9.5 PMM versus the normal version of MICE

Under MAR, MICE (PMM) shows a small downward bias and lower coverage than MICE (normal). Here the data really are linear and normal, so the normal model is exactly right and PMM's usual strength, being robust to unusual distributions, is not needed. A likely reason for its weakness is that PMM can only borrow values that were observed, and the rows that MAR targets (those with extreme predicted values) have few similar donors. PMM would be expected to do better with skewed or non-linear data, which this study does not contain.

9.6 MNAR is different in kind

Under MNAR the chance of being missing depends on the missing value itself, so the observed data cannot reveal the missingness process. The methods that assume MAR cannot correct for it, and none of the methods tried here does. When data may be MNAR, the honest approach is a **sensitivity analysis**: repeat the analysis under several plausible assumptions about the missing values and report how the conclusions change.

9.7 Why the results differ from an earlier one-variable version of this project

In an earlier version of this project only Y had missing values and the predictors were complete. In that setting complete case, regression imputation and EM give exactly the same coefficients, which hides their differences. In this version two variables (Y and X_1) have gaps and an auxiliary variable Z is available, so the methods really do behave differently. The sample size is also larger (200 instead of 100), and the experiment is repeated 300 times instead of being based on a single dataset.

10 Limitations

- **One simple model.** The data are linear and normal, the predictors are independent, n = 200 and about 20% of two variables are missing. Results can change with other distributions, sample sizes or amounts of missing data. In particular, PMM is designed for non-normal data, which this study does not contain.
- **Chosen strengths.** The strength of the MAR and MNAR rules (the number 2 in the formula) was chosen by us, and MNAR is only one of many possible forms.
- **A very helpful extra variable.** Z is a close stand-in for Y. Real auxiliary variables are usually weaker, which would shrink the advantage of MICE and EM over complete case under MAR.
- **Simplified methods.** MICE uses 5 cycles, 10 imputations, 5 donors and a normal (±1.96) interval rather than Rubin's t-based interval. EM assumes a normal distribution and, inside Part B, takes its standard error from only 40 bootstrap resamples. For real work use a tested package such as *mice* in R, *statsmodels* or scikit-learn's *IterativeImputer*.
- **Random error.** With 300 repetitions, coverage is accurate to about ±1.3 points. Part A is a single dataset and should not be used to rank methods.
- **Only β_1 is examined**, not the other coefficients or predictions.

11 Conclusion

How missing data are handled is part of the statistical analysis, not a detail to be skipped. It changes the estimate and it changes whether the confidence interval can be trusted. Table 13 summarises what works under each mechanism.

Mechanism	What works	What to avoid
MCAR	Complete case is acceptable. MICE or EM are slightly more precise.	Mean imputation and single imputation.
MAR	EM or MICE (normal), with the helpful extra variable in the imputation model.	Complete case (biased) and single imputation (over-confident).
MNAR	Nothing is reliable. Report a sensitivity analysis.	Claiming that any of these methods fixes MNAR.

Table 13: Recommendations by mechanism.

In every case:

- **Never use mean imputation** for inference.
- **A good point estimate is not enough.** Single imputation can be nearly unbiased and still badly over-confident. Carry the uncertainty through with multiple imputation (Rubin's rules), EM, or a bootstrap that repeats the filling-in.
- **Use all the information you have.** Include variables that explain the missingness or the outcome in the imputation model.

Appendix A: A Worked MICE Example

This appendix shows the numbers behind MICE (normal) for the three datasets of Part A. Each row lists the 10 estimates of β_1 (one per completed dataset), then the pooled estimate and the variances used by Rubin's rules.

Mechanism	m1	m2	m3	m4	m5	m6	m7	m8	m9	m10
MCAR	2.097	2.089	2.101	2.083	2.035	2.076	2.162	2.087	2.031	2.055
MAR	1.948	2.014	2.012	2.073	2.093	2.081	2.102	2.048	2.055	1.930
MNAR	2.087	2.132	2.070	2.148	2.104	2.120	2.092	2.123	2.089	2.101

Table A1: The 10 estimates of β1 from the 10 completed datasets (MICE, normal version).

Mechanism	Pooled β1	Within (W)	Between (B)	Total = W + 1.1 B	Pooled SE
MCAR	2.082	0.00511	0.00140	0.00665	0.082
MAR	2.036	0.00583	0.00349	0.00967	0.098
MNAR	2.107	0.00616	0.00057	0.00679	0.082

Table A2: Rubin's rules applied to the 10 estimates (m = 10, so 1 + 1/m = 1.1).

The within-imputation variance is the ordinary regression uncertainty inside one completed dataset. The between-imputation variance shows that different plausible fillings of the gaps lead to slightly different estimates. Single imputation has only the first part, which is why its standard error is too small.

Appendix B: Settings and How to Reproduce

Setting	Value
Observations per dataset	200
True model	Y = 5 + 2·X1 − 1.5·X2 + 1·X3 + ε, Z = Y + noise
Missing values	about 20% of Y and about 20% of X1
Part A	seed 42; 500 bootstrap samples per method
Part B	seed 2024; 300 repetitions; 40 bootstrap samples for EM's SE
Standard errors in Part B	naive OLS SE (complete case and single imputation); Rubin's rules (MICE); bootstrap (EM)
MICE	10 imputations, 5 cycles, PMM with 5 donors
Software	Python with numpy, pandas, scipy, scikit-learn, statsmodels, matplotlib
Run time	about 8 minutes for the full notebook

All code, the executed notebook, the figures and the result tables are in the project repository (see the README). Every code block in the notebook starts from its own random seed, so any block can be re-run on its own and gives the same answer.

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